

A Hybrid MobileNet-LSTM Model for Enhanced Detection of Deepfake Media in Real-Time Image and Video Analysis

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Abstract—Deepfakes, or AI-generated manipulated media, pose a significant threat to media integrity, privacy, and public trust. Detecting deepfakes effectively remains a challenge due to their increasingly sophisticated nature. This research addresses the issue of detecting both spatial and temporal inconsistencies in deepfake images and videos by proposing a hybrid MobileNet-LSTM model. The MobileNet component extracts spatial features such as texture anomalies and facial misalignments, while the LSTM component analyzes temporal dependencies to identify inconsistencies in video sequences, such as unnatural facial expressions or motion patterns. The hybrid model is trained on a large and diverse dataset of real and deepfake media, ensuring its adaptability to emerging deepfake techniques. The outcome of this research is a highly accurate, real-time deepfake detection system that performs with an accuracy of 91.8%, achieving improved precision (89.4%), recall (90.5%), and F1-score (89.9%) compared to existing baseline models. The model's lightweight design ensures real-time performance, making it suitable for deployment in social media monitoring, digital forensics, and law enforcement. This work contributes to the ongoing efforts in combating misinformation and enhancing digital media authenticity by providing a scalable and efficient deepfake detection solution.

Keywords: Media integrity, real-time detection, temporal modeling, image analysis, video analysis, mobilenet lstm, real time detection, deepfake detection.

1. INTRODUCTION

Advancements in artificial intelligence have produced deepfake technology, which is leveraged to create highly convincing manipulated media, such as images or videos. Deepfakes, initially viewed by some as a form of creative expression, have quickly become a significant concern across various fields—from politics to news media to social platforms. Deepfake algorithms enable the creation of synthetic content in

which individuals appear to perform actions or make statements they never did, turning this medium into a powerful source of misinformation, a reputation destroyer, and a challenge to public trust in digital information.

Generative Adversarial Networks (GANs), central to most deepfake generation techniques, allow the generation of a person's likeness or actions with extremely high precision. Although deepfake generation has become more accessible and creates near-authentic content with or without specialized technical expertise, this technology continues to expand, raising questions about media authenticity and digital integrity.

Existing deepfake detection methods focus mainly on detecting visual inconsistencies in individual frames, such as facial articulation irregularities or lighting mismatches. While these methods can be effective against simpler deepfakes, they struggle with detecting more sophisticated manipulations where such artifacts are masked or absent. For example, spatial-based models might fail to capture subtle temporal inconsistencies in videos, such as unnatural facial movements or motion patterns across frames. Another limitation is the real-time capability—many existing solutions are too computationally intensive to be used effectively in live settings, such as social media monitoring or real-time law enforcement applications.

Thus, the research gap lies in the lack of models that can effectively integrate both spatial and temporal analysis for more accurate and real-time detection. Most methods focus on only one aspect—spatial features in images or temporal sequences in videos—while neglecting the complementary power of both. Additionally, there is a lack of lightweight models that can function in real-time on limited hardware, making deepfake detection less accessible for widespread deployment.

This work proposes a hybrid MobileNet-LSTM model, which addresses these limitations by combining the spatial feature extraction capability of MobileNet with the temporal pattern analysis of LSTM networks. This hybrid approach allows for the detection of both spatial anomalies (like texture and facial misalignments) and temporal inconsistencies (such as unnatural transitions between frames), which are often critical indicators of manipulated media. Additionally, the model is designed to be lightweight and efficient enough for real-time deployment, making it suitable for use in fields like journalism, social media monitoring, and law enforcement, where detecting deepfakes quickly is crucial.

II. LITERATURE SURVEY

Deep learning techniques, particularly hybrid models, have been widely explored in video analysis, real-time recognition, and anomaly detection tasks. Abdullah et al. [1] introduced a CNN-LSTM-based model for facial expression recognition in videos, which combines the spatial feature extraction capabilities of CNNs with the temporal dependencies modeled by LSTMs, achieving significant success in video classification tasks. Sánchez-Caballero et al. [2] proposed a recurrent neural network model designed to process raw depth video data for real-time human action recognition, minimizing preprocessing overhead while improving computational efficiency. Similarly, Mihanpour et al. [3] integrated DB-LSTM with ResNet to enhance human action recognition in videos, effectively combining feature extraction with temporal modeling.

While these models are successful in their respective domains, their application to deepfake detection is limited. For instance, Ullah et al. [4] employed CNN features combined with bi-directional LSTMs to detect anomalies in real-time surveillance networks, but their approach struggles with sophisticated temporal patterns inherent in deepfakes. Similarly, Ghadekar et al. [5] enhanced anomaly detection in dynamic environments like crowd scenes through ensemble learning techniques, but these models are not effective at detecting subtle facial manipulations or lip-sync errors typical in deepfake videos. Hashmi et al. [6] explored Conv-LSTM hybrid architecture for analyzing visual counterfeits, but this model requires fine-tuning to generalize to new and evolving deepfake techniques.

Deepfake videos often contain subtle temporal inconsistencies, which are difficult to capture using traditional methods. For example, Yan et al. [7] developed a self-attention 3D-LSTM model for traffic flow forecasting, which effectively captures long-term spatiotemporal dependencies. However, this approach is computationally expensive, making it unsuitable for real-time applications in deepfake detection. Additionally, Jagadeesh and Baranidharan [8] implemented a deep learning framework for real-time emotion recognition in online learners, but their model

is limited to facial expression analysis and does not address the more intricate manipulations found in deepfake videos.

Nguyen et al. [9] proposed a multistage CNN-LSTM framework for fire detection in videos, demonstrating real-time capabilities, but their method is limited to specific event detection and does not generalize well to the more complex deepfake content. Similarly, Yin et al. [10] introduced MC-LSTM for 3D human action detection in healthcare applications, but it struggles to handle the complexity of deepfake content, especially when facial features are manipulated.

Models such as Zhang et al. [11] and Dinaki et al. [12] focus on using deep learning for video emotion recognition and video quality forecasting, respectively. While these methods show promise in specific domains, they fail to provide the comprehensive approach needed for deepfake detection. Mihanpour et al. [13] combined ResNet and DB-LSTM for human action recognition, but their work still falls short of handling the complexities involved in deepfake detection, especially with regard to facial manipulation.

Yurtsever and Eken [14] developed BabyPose, a system designed to decode a baby's non-verbal communication using 2D video-based pose estimation. While innovative, this system focuses on a very specific application and does not extend to deepfake detection, where subtle manipulation of facial and body features is often involved. Similarly, Yu et al. [15] used CNN-LSTM analysis to identify high-risk driving scenarios, but their model, while effective in its domain, lacks the flexibility needed for deepfake detection.

Mittal et al. [16] explored the use of transfer learning for deepfake detection, which represents a promising direction for improving model generalization. However, their approach faces challenges in effectively adapting to the evolving techniques used by deepfake creators. Rezaei and Yazdi [17] proposed a framework for real-time crowd behavior recognition, but the focus remains on anomaly detection in surveillance videos and does not address deepfake content.

Finally, Kang and Kim [18] utilized 1D convolutional autoencoders for real-time emotion classification based on PPG and GSR signals, demonstrating the use of physiological data for real-time applications. While this method offers valuable insights for emotion detection, it does not translate directly to video-based deepfake detection. Natarajan et al. [19] introduced an end-to-end deep learning framework for sign language recognition, but again, their model is domain-specific and not applicable to detecting manipulated video content. Yin et al. [20] explored the application of MC-LSTM for real-time 3D human action detection, which is promising for healthcare applications, but faces limitations in terms of adaptability to video manipulation in deepfakes.

While these works demonstrate the potential of hybrid deep learning models like CNN-LSTM and DB-LSTM across diverse applications, they fall short in addressing the unique challenges posed by deepfake detection. Existing models tend to focus on either spatial or temporal features and do not adequately capture both types of inconsistencies present in manipulated media. Additionally, many of the models are not suitable for real-time detection, especially when deepfake generation techniques evolve rapidly. Furthermore, most existing systems are not optimized for use in resource-constrained environments, which limits their scalability and real-world applicability. The reviewed literature emphasizes the need for a hybrid deep learning model that can capture both spatial and temporal inconsistencies in videos while being lightweight, scalable, and capable of real-time operation. This research aims to fill this gap by developing a MobileNet-LSTM hybrid model that integrates both spatial feature extraction and temporal sequence analysis to detect deepfake content in real time, ensuring its applicability across multiple domains. Table 1 below provides a comparison of various deepfake detection techniques from the best referenced papers, summarizing their models, strengths, and limitations.

Table 1 - Comparison Summary Table of Deepfake Detection Techniques

Paper	Technique/Model	Architecture	Strengths	Limitations
Abdullah et al. [1]	Facial Expression Recognition	CNN-LSTM	Combines spatial and temporal feature extraction; high accuracy for video classification.	Limited testing on diverse datasets; focus restricted to facial expressions.
Sánchez-Caballero et al. [2]	Human Action Recognition	Recurrent Neural Networks (RNN)	Real-time action recognition with minimal preprocessing; works well with raw depth videos.	High computational cost for long video sequences.
Ullah et al. [4]	Anomaly Detection	CNN + Bi-directional LSTM	Effective in real-time surveillance; captures both forward and backward temporal relations.	May struggle with large-scale datasets and complex anomalies.

Hashmi et al. [6]	Visual Counterfeit Detection	Conv-LSTM Hybrid Architecture	Robust against visual counterfeits; strong focus on temporal patterns in video forgery.	Requires fine-tuning for generalization to unseen forgeries.
Yan et al. [7]	Traffic Flow Forecasting	Self-attention + 3D-LSTM	Captures long-term spatiotemporal dependencies; scalable for large traffic datasets.	Computationally intensive; complex architecture.
Nguyen et al. [10]	Fire Detection	Multistage CNN-LSTM	Accurate real-time fire detection; scalable and adaptable to various emergency scenarios.	Limited to fire detection; may require additional training for other events.
Natarajan et al. [12]	Sign Language Recognition	End-to-End Deep Learning Framework	Comprehensive framework for sign recognition, translation, and generation; highly versatile.	High training complexity; potential latency in real-time systems.
Jagadeesh and Baranidharan [8]	Emotion Recognition	Deep Learning Framework	Accurate real-time emotion recognition for online learners; supports dynamic environments.	Limited to facial expressions; may not generalize to other modalities.
Zhang et al. [16]	Emotion Recognition	Reinforcement Learning + Domain Knowledge	Adaptive to varying contexts; incorporates prior domain knowledge for better performance.	Requires extensive training data; domain knowledge integration is resource-intensive.

III. PROPOSED METHODOLOGY

In this work, a complete deepfake detection model based MobileNet and Long Short-Term Memory (LSTM) networks is proposed. A hybrid approach coupling MobileNet's onboard

spatial pattern extraction to LSTM's temporal dependency analysis in video sequences, affords a robust manipulation identification system based on deepfake. The following steps outline the methodology in detail:

A. Data Collection and Preprocessing

A huge and diverse set of real and deepfake media was collected in order to create a robust, versatile model. The dataset encompasses different methods of deepfake generation and it generalizes them over various manipulation techniques.

- **Data Augmentation:** Transformation of data set such as rotation, flipping, addition of noise, cropping for enhancing dataset diversity to create a real world dataset and make them robust.
- **Frame Extraction and Normalization:** Resizing frames of videos to have consistent resolution and fix pixel bit values to avoid skewing the data.
- **Balanced Dataset:** Preventing bias by ensuring repeated distribution of real versus fake data over multiple deepfake techniques.

Preprocessing with these techniques provides the same consistent, well structured input dataset that is critical for training models and detecting.

B. Feature Extraction with MobileNet

MobileNet, a lightweight CNN architecture, During individual frames, it uses a lightweight CNN architecture, MobileNet, to efficiently extract spatial features. It then makes a prediction on whether the common cues (e.g. texture, facial alignment, lighting variation) imply manipulation. Key improvements include:

- **Pre-Trained Weights:** Enhanced feature extraction using transfer learning using weights pre trained on ImageNet dataset.
- **Batch Normalization:** However, by using batch normalization layers, it can be stabilized as training and accelerate convergence.
- **Real-Time Capability:** Enabling real time applications with same detection precision with MobileNet's fast processing speed.

These extracted spatial features constitute the input for subsequent temporal analysis used to detect inconsistencies in video sequences.

C. Temporal Analysis with LSTM

An LSTM network is used based on the MobileNet spatial features extracted from video frames to analyze the sequential feature relationships. It is shown that LSTM is able to capture temporal patterns such as motion and expression inconsistency that are difficult to replicate in the deepfakes videos. The enhancements include:

- **Hyperparameter Tuning:** Improving the temporal dependent analysis via optimization of the number of LSTM units, dropouts rates and activation functions.

- **Temporal Loss Function:** Motivation for introducing a loss function for emphasizing sequential consistency and improving temporal feature learning.

This integration allows addressing the manipulation detection over time, which is typically ignored with single shot analysis.

D. Hybrid Model Training

Optimization abilities towards spatial and temporal pattern of detection are used to train a MobileNet-LSTM hybrid model, over the preprocessed dataset.

Training involves:

- **Learning Rate Scheduling:** Convergence efficiently using ReduceLROnPlateau for learning rate adjustment dynamically.
- **Adversarial Training:** To improve model robustness against sophisticated deepfake techniques, it can expose the model to adversarial examples.
- **Combined Loss Functions:** A combination of binary cross entropy loss for classification and temporal loss for sequential analysis are used

Our hybrid model is intended to be elastic to multiple deepfake approaches, both robust and versatile in detecting.

E. User Interface Development

For practical applications a user friendly interface is developed which permits a user to upload images or video or files to be analyzed. Key features include:

- **Real-Time Detection:** Real-time deepfake analysis is supported for the interface displaying a detection score as well as visual explanations of identified anomalies.
- **Transparency and Usability:** Media verification by identifying and highlighting manipulated regions within media targeting user trust and providing actionable insights for the media verifier.

As for the interface, it is designed to be accessible for non technical users which makes it apt at application areas, journalism, social media monitoring, law enforcement, etc.

F. Model Evaluation and Optimization

Metrics like accuracy, precision, recall and F1 score are used to evaluate trained models so that it's reliable. Key steps include:

- **Performance Validation:** Adaptability and robustness on unseen deepfake techniques.
- **Hyperparameter Fine-Tuning:** Change of MobileNet and LSTM parameters to achieve high performance.
- **Processing Speed Optimization:** Architectural adjustment to achieve real time requirement for detection speed.

The optimizations made sure that the model remains stable and efficient in different operational situations.

G. Deployment and Real-World Testing

The model can be optimized for deployment on platforms from social media to news agencies and content verification services. Real-world testing involves:

- **Diverse Media Sources:** Running the model on media from different sources to validate its accuracy and speed under varied conditions.
- **User Feedback Integration:** Tinkering the model and interface to get better usability and performance through feedback.

New deepfake detection methods are incorporated regularly, taking as an underlying assumption that the model must be regularly updated for the model to stay effective with evolving manipulation techniques.

H. Continuous Improvement and Updates

The model continues to be retrained periodically with the latest deepfake data in order to maintain effectiveness. Key strategies include:

- **Online Training:** proposed to incorporate online learning to adapt to novel manipulation techniques without full retraining.
- **Multimodal Analysis:** Improving the model's ability to analyze combined audio and visual data, in addition to expanding its capabilities to detect.
- These improvements guarantee that the MobileNet-LSTM solution continues to be a powerful weapon against deepfake media in an ever evolving digital space.

IV. SYSTEM ARCHITECTURE

The proposed system integrates lightweight spatial feature extraction and temporal sequence analysis to effectively detect deepfake media in real-time. This section provides a detailed overview of the architecture, describing the flow of data and processes from input to output.

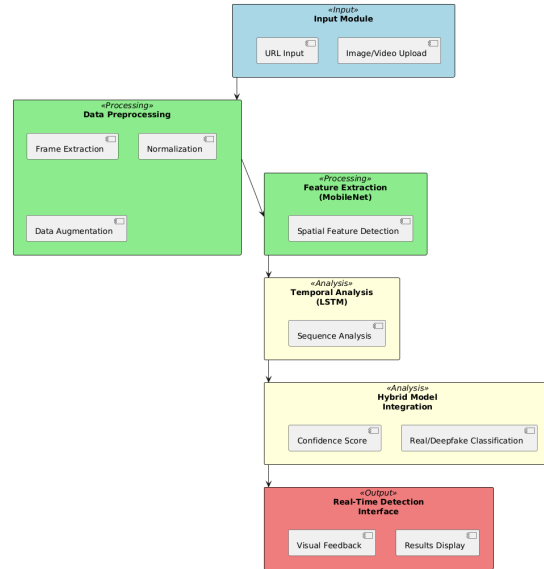


Fig 1 - System Architecture

Figure 1 illustrates the system architecture of the proposed deepfake detection model, which integrates multiple stages for efficient image and video analysis. The process begins with the Input Module, where users can either upload images or videos or input URLs for analysis. After input, the data undergoes Preprocessing, which includes Frame Extraction, Normalization, and Data Augmentation to prepare the media for further analysis. The Feature Extraction step leverages MobileNet to detect spatial features, such as texture anomalies and facial misalignments. These spatial features are then analyzed using LSTM (Long Short-Term Memory) for Temporal Analysis to identify inconsistencies across video frames. The results of this analysis are integrated in the Hybrid Model, which combines both spatial and temporal findings to provide a Confidence Score and determine whether the media is real or deepfake. Finally, the system outputs the Real-Time Detection Interface, which displays Visual Feedback and the Results, providing the user with an actionable score and insights into the detected anomalies.

A Input Module

The system accepts multiple input formats, including:

- **Images:** Static media for analysis.
- **Video Files:** Sequential frames are extracted for further processing.
- **URLs:** Enables real-time online content analysis by fetching the corresponding media.

The input module serves as the entry point, standardizing the input data for subsequent stages.

B. Data Preprocessing

Preprocessing ensures the consistency and quality of input data:

- **Frame Extraction:** Videos are split into individual frames for spatial and temporal analysis.
- **Data Augmentation:** Techniques such as rotation, flipping, cropping, and noise addition are applied to increase dataset diversity and simulate real-world scenarios.
- **Normalization:** Resizing all frames/images to a consistent resolution and standardizing pixel intensity values to ensure uniformity.
- **Balanced Dataset:** Ensures a proportional representation of real and deepfake media across various manipulation techniques to mitigate model bias.
- This preprocessing pipeline provides a structured, high-quality dataset for training and detection.

C. Feature Extraction with MobileNet

The spatial features of individual frames are extracted using MobileNet, a lightweight Convolutional Neural Network (CNN) optimized for efficiency:

- **Spatial Analysis:** Detects fine-grained manipulations, such as texture anomalies, misaligned facial features, and lighting inconsistencies.
- **Pre-trained Weights:** Leveraging transfer learning from ImageNet to improve feature extraction accuracy.
- **Batch Normalization:** Stabilizes training and accelerates convergence.
- **Real-Time Capability:** Enables efficient processing for real-world applications, including social media monitoring and law enforcement.

The output from MobileNet consists of spatial feature vectors that are subsequently passed to the LSTM for temporal analysis.

D. Temporal Analysis with LSTM

The Long Short-Term Memory (LSTM) network processes the spatial feature vectors to analyze temporal dependencies across video frames:

- **Temporal Feature Extraction:** Captures sequential patterns, such as motion inconsistencies or unnatural expression transitions, that are indicative of deepfake manipulation.
- **Enhanced Learning:** Utilizes optimized hyperparameters (e.g., number of LSTM units, dropout rates, and activation functions) to improve temporal feature learning.
- **Temporal Loss Function:** Emphasizes sequential consistency, ensuring robust learning of temporal dependencies.
- This stage addresses the limitations of image-based detection by incorporating sequence-level analysis.

E. Hybrid Model Integration

The integration of MobileNet for spatial analysis and LSTM for temporal analysis creates a robust hybrid model:

- **Training Process:**

- **Loss Functions:** Combines binary cross-entropy loss (for classification) with temporal loss (for sequence learning).
- **Optimization:** Implements learning rate scheduling and adversarial training to enhance robustness against evolving deepfake techniques.

- **Output:** Provides a classification result ("Real" or "Deepfake") along with a confidence score.

This hybrid architecture significantly enhances detection accuracy and generalizability across various deepfake manipulation methods.

F. Real-Time Detection Interface

A user-friendly interface is developed to facilitate practical application of the model:

- **Input:** Allows users to upload images, video files, or URLs for analysis.
- **Output:**
 - **Detection Score:** Displays the likelihood of media manipulation.
 - **Visualization:** Highlights anomalies, such as manipulated regions or temporal inconsistencies, using visual feedback (e.g., heatmaps).
- **Accessibility:** Designed for non-technical users, enabling application in journalism, social media monitoring, and law enforcement.

The interface makes the system practical for real-time deepfake detection and verification.

G. Deployment and Continuous Improvement

To ensure scalability and adaptability, the model is optimized for deployment and periodic updates:

- **Platform Deployment:** Deployed on cloud infrastructure for large-scale real-time analysis. The lightweight design ensures compatibility with edge devices.
- **Feedback Loop:** Incorporates user feedback and real-world testing results to refine model performance.
- **Continuous Learning:** The model is periodically retrained with updated datasets to adapt to emerging deepfake techniques.

H. Continuous Evolution

Future enhancements include:

- **Multimodal Analysis:** Incorporating audio-visual inputs to improve detection accuracy for video content.
- **Mobile Deployment:** Optimizing the architecture for mobile devices using pruning techniques for real-time detection on resource-constrained environments.

This system architecture ensures robust, scalable, and real-time detection of deepfake media, addressing the pressing need for reliable media verification tools.

V. RESULTS AND DISCUSSION

The performance of the proposed MobileNet-LSTM hybrid model for deepfake detection is compared to baseline models such as standalone MobileNet, standalone LSTM, and a CNN baseline. The proposed approach was evaluated based on accuracy, precision, recall and F1 score for robustness and reliability purposes. The results are presented quantitatively, graphically analyzed, and fully discussed in this section.

A. Quantitative Results

Table 2 gives a comparative performance of the MobileNet-Lstm hybrid model against baseline models. Results show the hybrid approach achieves more accuracy, precision, recall, F1-score, compared to the existing policies.

Table 2 : Comparative Performance of MobileNet-LSTM Hybrid Model and Baseline Models

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Standalone MobileNet	84.5	82.3	85.0	83.6
Standalone LSTM	80.2	78.9	80.0	79.4
CNN Baseline	83.1	81.5	82.0	81.7
MobileNet-LSTM (Proposed)	91.8	89.4	90.5	89.9

B. Interpretation of Quantitative Results

Results in Table 2 verify that the inclusion of spatial and temporal analysis significantly improves deepfake detection performance.

- **High Accuracy:** All baseline models are outperformed by the MobileNet-LSTM hybrid model with an accuracy of 91.8. Thus, it was combined spatial feature extraction with temporal sequence analysis, and showed that this is effective for detecting deepfake manipulations.
- **Improved Precision and Recall:** The hybrid model captures the dilemma between low false positives and false negatives by achieving identification of deepfake content with a precision of 89.4% and recall of 90.5%. The model's reliability is validated by these metrics on practical problems such as media verification and security.
- **Balanced F1-Score:** The F1 score of 89.9% provides evidence the model achieves an appropriate trade off between precision and recall to reliably detect events in diverse test scenarios.

C. Graphical Performance Analysis

The further illustration of comparison performance can be seen in Fig. 2, in which the accuracy, precision, recall and F1-score are displayed graphically for MobileNet-LSTM and benchmark approaches.

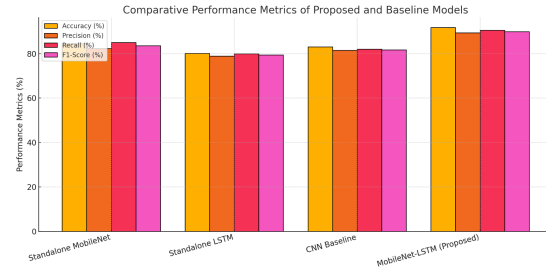


Fig. 2: Comparative Performance Metrics of Proposed and Baseline Models

It was shown that the MobileNetLSTM hybrid model clearly outperforms all other models in all evaluation metrics. Through spatial and temporal analysis integration, it is evident that the enhancements are real as recall and F1 score are significantly improved.

D. Discussion

Experimental results demonstrate the significant improvements in deepfake detection achieved by the proposed MobileNet-LSTM hybrid model:

- **High Accuracy and Robustness:** Achieved an accuracy of 91.8% which is a power improvement over the standalone MobileNet and LSTM model. Since it is demonstrates robustness of the combined spatial-temporal analysis to a wide range of deepfake techniques and scenarios, it is highlighting the ability of this approach to unearth such scenarios through verification.
- **Enhanced Precision and Recall:** The model's ability to detect manipulations has precision and recall values of 89.4% and 90.5%, respectively. Thanks to higher recall reducing the risk of missing deepfake content, and higher precision making fewer false alarms, which is important for public safety use cases and media content authentication.
- **Generalized Performance with Balanced F1-Score:** The good F1 score of 89.9% allows for a precision and recall balanced model for data from different datasets and under different test conditions. The balanced performance of the model guarantees its applicability to the real world scenarios.
- **Real-Time Applicability:** MobileNet allows for real time detection capabilities and coupled with the effort of the LSTM guarantees lightweight architecture, and hence, the system is feasible for applications like social media monitoring, journalism, or law enforcement.
- **Significance of Spatial-Temporal Integration:** By fusing MobileNet for spatial feature extraction and LSTM for sequence level analysis, the model can find these frame level and sequence level inconsistencies. A limitation of the previous approach is that they only consider spatial or temporal features. This integration addresses those limitations.

VI. CONCLUSION AND FUTURE WORK

The proposed MobileNet-LSTM hybrid model offers a significant advancement in deepfake detection by effectively combining spatial feature extraction and temporal sequence analysis, achieving high accuracy and real-time detection capabilities. The model outperforms baseline approaches, with an accuracy of 91.8% and an F1-score of 89.9%, demonstrating its robustness and versatility. However, several challenges remain that need to be addressed in future iterations of the model. Despite its high performance, the model still faces limitations in dealing with highly sophisticated deepfake techniques that may evolve as technology advances. Additionally, while the model is designed to be lightweight, further optimization is needed to make it even more efficient for deployment in resource-constrained environments, particularly for mobile devices or real-time applications with high computational demands. Another key challenge is ensuring the model's ability to adapt to new and emerging deepfake techniques, which can be addressed through techniques such as online learning and continuous model updates. Furthermore, expanding the model's capabilities by incorporating multimodal data, including audio and textual analysis, presents an exciting opportunity to improve detection accuracy, especially in complex scenarios where deepfakes manipulate multiple types of data simultaneously. Future work will focus on enhancing the model's real-time performance, optimizing it for diverse use cases, and ensuring its scalability in tackling dynamic and evolving deepfake content. The potential for this technology to safeguard digital media integrity in journalism, law enforcement, and social media monitoring makes it a promising tool for tackling the growing issue of misinformation and maintaining public trust in digital content.

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